Geometric Prediction and Exact-Root Certification of Finite-Error Ordering in Quantum Control

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Abstract

Quantum controls that agree at the nominal endpoint and in their complete first-order response to the declared global error coordinates need not have the same finite-error robustness. We study this residual implementation dependence in a serialized two-atom Rydberg model—a minimal interacting setting in which amplitude, detuning, and interaction-calibration responses coexist—with twenty-four control phases and three global error coordinates.

The main result is a computer-assisted exact-root finite-error ordering theorem. For a frozen twelve-path cohort, strict 192-bit Krawczyk inclusions certify twelve locally unique roots whose projective output states and first projective responses exactly match the reference. Direct outward-rounded Arb propagation of the certified root boxes through the declared six-axis error functional produces twelve pairwise disjoint performance intervals, totally ordered according to an order frozen before the propagation, certifying the complete finite-error ordering for all 66 unordered path pairs. Two computationally independent mechanism certificates expose the internal structure of this ordering: a zero-point Taylor jet through order thirty, with formal Cauchy-alias and Dyson-tail enclosures, certifies 52/66 comparisons with no reversed pair, while the lowest-order quartic term alone certifies 34/66.

On a separate prospective cohort, a frozen quartic response contraction strongly predicts the held-out finite-error ranking (Spearman $\rho = 0.996992$ on twenty paths), and its tensor representation is verified to transform covariantly under nonorthogonal error-coordinate changes. This prospective evidence is not a premise of the theorem. The theorem is conditional on the declared finite-dimensional model and error functional; it does not constitute hardware, model-discrepancy, global-fibre, or many-body evidence.

1 Introduction

Programmable neutral-atom systems expose a rich pulse-level control space, including Rabi amplitude, phase, detuning, atomic geometry, and Rydberg-mediated interactions [1, 2, 3]. This flexibility permits many control schedules to realize the same nominal task. It also creates an identification problem: nominally equivalent implementations can respond differently when calibration errors are finite.

Endpoint equivalence does not resolve this ambiguity. Even matching the complete first-order output-state response to the relevant error coordinates may leave a family of implementations with different higher-order behaviour. The resulting question is not whether leading errors can be cancelled, but whether the remaining finite-error ordering can be inferred before those finite-error outcomes are evaluated.

Composite-pulse and dynamically-corrected-gate constructions suppress designated low-order error terms [4, 5, 6, 15, 16, 18]; filter-function methods characterize the frequency-domain noise susceptibility of a fixed control [17]; quantum geometric approaches endow state and parameter manifolds with the quantum geometric tensor and its curvature [19]; and control-landscape theory analyses the critical-point structure of control objectives [13, 14]. These methods typically optimize or cancel designated low-order sensitivities of a control. What remains unresolved by these viewpoints is whether controls that are exactly indistinguishable at the endpoint and throughout their complete first-order projective response can nevertheless admit a rigorously certifiable finite-error ordering. Here we condition on that strong equivalence relation—candidate controls must share both the nominal output state and the complete first-order output-state response to global amplitude, detuning, and interaction errors—and study the finite-radius ordering that remains on the implementation fibre, together with its rigorous certification. We then ask:

> *Within the remaining implementation fibre, can a zero-error geometric quantity prospectively predict finite-error robustness, and can that ordering be certified at locally unique projectively matched controls?*

This formulation separates two logically distinct tasks. The first is prediction: a local quantity must be fixed before held-out finite-error evaluation and must rank previously unseen paths. The second is certification: numerical matching must be replaced by exact-root existence and uniqueness, and the finite-error ordering must be enclosed by outward-rounded interval arithmetic, in the validated-numerics tradition of computer-assisted proof [11, 10, 12]. High correlation alone does not provide the second conclusion.

We study a two-atom Rydberg model with twenty-four piecewise-constant phase variables. The numerical matching Jacobian has rank sixteen, leaving an eight-dimensional numerical tangent null space that organizes the candidate implementation fibre.

1.1 Main results

Main theorem (informal statement). Consider the frozen twelve-path cohort, the serialized finite-dimensional two-atom Hamiltonian, the declared transverse phase boxes, and the six-axis mean finite-error functional. For each declared control, a strict Krawczyk inclusion certifies a locally unique root exactly matched in projective output state and first projective response. Direct 192-bit outward-rounded Arb propagation over the twelve certified root boxes produces pairwise disjoint performance intervals whose orientations agree with the pre-outcome frozen ordering for all $\binom{12}{2} = 66$ unordered path pairs. The formal statements and computer-assisted proofs are given in Theorems 1 and 2. This result is conditional on the serialized model and declared error functional; it is not a claim about hardware, model discrepancy, global fibre topology, or many-body scaling.

Mechanism proposition (informal statement). Over the same certified Krawczyk boxes, propagation of the zero-point Taylor jet through order thirty, including formal Cauchy-alias and analytic tail enclosures, certifies 52/66 pairwise comparisons with no reversed certified pair. This order-thirty calculation is a separate, computationally distinct mechanism certificate. It is not a premise or computational step in the direct proof of the main theorem. The formal statement appears in Proposition 2.

Reading guide. Four quantitative results play different logical roles:

66/66	direct exact-root theorem (main result)
52/66	order-thirty mechanism certificate
34/66	quartic-only boundary
$\rho = 0.996992$	prospective evidence (separate cohort)

The frozen order used in the exact-root theorem is supplied by the pre-outcome order-thirty certificate for the twelve-path cohort; no pathwise identity between the twenty-path prospective cohort and the twelve-path formal cohort is claimed.

The remainder of the paper follows the logical development from prospective prediction to exact-root certification. Section 2 fixes the model and the benchmark error law; Section 3 defines exact projective response matching and proves a model-independent quadratic-degeneracy lemma; Section 4 introduces the prospective quartic predictor; Section 5 constructs its covariant tensor form; and Section 6 certifies locally unique exact projective matched roots and proves the direct finite-error ordering theorem, together with the separate order-thirty mechanism certificate.

The result is intentionally model-conditional. It concerns the serialized finite-dimensional two-atom Hamiltonian, the declared transverse boxes, and the six-axis mean finite-error functional. It does not establish global fibre topology, worst-case ranking, many-body scaling, open-system robustness, or hardware-level performance.

2 Two-atom control model

2.1 Hamiltonian

We use the ordered basis

$$\{|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle\}.$$

Let

$$X_\Sigma = X \otimes I + I \otimes X, \qquad Y_\Sigma = Y \otimes I + I \otimes Y,$$

$$N_\Sigma = n \otimes I + I \otimes n, \qquad N_{rr} = n \otimes n,$$

where $n = |r\rangle\langle r|$. During segment j, the Hamiltonian is

$$H_j(\epsilon) = \frac{\Omega}{2}(1+\epsilon_\Omega)\big(\cos\phi_j\, X_\Sigma + \sin\phi_j\, Y_\Sigma\big) - \Omega\,\epsilon_\Delta\, N_\Sigma + V(1+\epsilon_V)\, N_{rr}. \tag{1}$$

The dimensionless error vector is

$$\epsilon = (\epsilon_\Omega, \epsilon_\Delta, \epsilon_V).$$

The nominal parameters are

$$\Omega = 2\pi \text{ rad}/\mu\text{s}, \qquad V = 4\Omega,$$

with 24 segments of duration

$$\tau = 0.1\ \mu\text{s}, \qquad T = 24\tau = 2.4\ \mu\text{s}.$$

The nominal interaction is represented as $V = C_6/R^6$. Using the frozen Pulser 1.8.0 `DigitalAnalogDevice` value $C_6 = 5420158.53$ rad μm$^6/\mu$s and the declared $V = 4\Omega = 8\pi$ rad$/\mu$s gives

$$R = (C_6/V)^{1/6} = 7.74394075\ldots\ \mu\text{m}.$$

This distance is a serialized model parameter rather than a hardware-calibrated measurement.

The two-atom model is not intended as a many-body approximation. It is a minimal interacting Rydberg setting in which drive-amplitude, detuning, and interaction-calibration errors coexist. A single-atom model has no nontrivial N_{rr} response and therefore cannot represent the three-direction matched response problem studied here.

For a control path

$$\gamma = (\phi_1, \ldots, \phi_{24}) \in \mathbb{T}^{24},$$

the output state is

$$|\psi_\gamma(\epsilon)\rangle = U_{24}(\epsilon)\cdots U_2(\epsilon)U_1(\epsilon)|gg\rangle, \qquad U_j(\epsilon) = \exp[-i\tau H_j(\epsilon)], \tag{2}$$

with time-ordered application from segment 1 to segment 24. All primary calculations use exact matrix exponentials for the piecewise constant, finite-dimensional model. A pulse-level emulator was used in an earlier two-atom cross-check, but the formal certificate in this work is independent of that emulator and does not invoke a cloud backend. Throughout, *formal certificate* denotes an outward-rounded interval certificate computed in Arb/ACB ball arithmetic, not a proof-assistant formalization.

2.2 Loss and held-out error law

Let γ_{ref} denote the reference schedule and define

$$|\psi_{\mathrm{ref}}\rangle = |\psi_{\gamma_{\mathrm{ref}}}(0)\rangle.$$

Two infidelities are used. For the extraction of local response coefficients, the target is the path-local nominal state,

$$\mathcal{I}_\gamma^{\mathrm{loc}}(\epsilon) = 1 - |\langle\psi_\gamma(0)|\psi_\gamma(\epsilon)\rangle|^2. \tag{3}$$

For held-out and direct finite-error comparisons, the target is the common reference state,

$$\mathcal{I}_\gamma^{\mathrm{ref}}(\epsilon) = 1 - |\langle\psi_{\mathrm{ref}}|\psi_\gamma(\epsilon)\rangle|^2. \tag{4}$$

At an exact response-matched root, $[\psi_\gamma(0)] = [\psi_{\mathrm{ref}}]$ projectively, so the two definitions coincide; for numerically matched prospective cohorts, they differ only within the declared matching tolerance. Unless stated otherwise, $\mathcal{I}_\gamma^{\mathrm{ref}}$ below denotes (4).

The held-out error design contains the six signed axial errors

$$E = \{\pm 0.06\, e_\Omega,\ \pm 0.04\, e_\Delta,\ \pm 0.05\, e_V\}. \tag{5}$$

These axis amplitudes define a dimensionless benchmark error law for certificate comparison; they are not inferred from hardware calibration data. The primary performance functional is the mean

$$\bar{\mathcal{I}}_\gamma = \frac{1}{6}\sum_{\epsilon\in E}\mathcal{I}_\gamma^{\mathrm{ref}}(\epsilon). \tag{6}$$

Worst-case infidelity is retained as a secondary diagnostic. The present predictors do not reliably rank the worst case, and no worst-case theorem is claimed.

3 Matched implementation fibre

3.1 Endpoint and first-order constraints

At zero error, remove the unobservable global phase by the path-local horizontal projector

$$P_{\gamma,\perp} = I - |\psi_\gamma(0)\rangle\langle\psi_\gamma(0)|,$$

and define the path-local horizontal first-order responses

$$J_{\gamma,a} = P_{\gamma,\perp}\frac{\partial}{\partial\epsilon_a}|\psi_\gamma(\epsilon)\rangle\Big|_{\epsilon=0}, \qquad a\in\{\Omega,\Delta,V\}. \tag{7}$$

Since $\psi_\gamma(0)$ and ψ_{ref} agree only up to a global phase, align them before comparison by the common phase factor

$$c_\gamma = \frac{\langle \psi_{\mathrm{ref}} | \psi_\gamma(0) \rangle}{|\langle \psi_{\mathrm{ref}} | \psi_\gamma(0) \rangle|} \in U(1), \qquad \widetilde{J}_{\gamma,a} = c_\gamma J_{\gamma,a}, \tag{8}$$

and compare the phase-aligned response $\widetilde{J}_{\gamma,a}$ with $J_{\mathrm{ref},a}$.

Definition 1 (Numerically matched implementation fibre)**.** For a declared tolerance η_F, the numerical matched fibre $\mathcal{F}_\eta$ consists of controls γ satisfying

$$\bar{\mathcal{I}}_\gamma(0) \le \eta, \qquad \|\widetilde{J}_{\gamma,a} - J_{\mathrm{ref},a}\| \le \eta, \quad a \in \{\Omega, \Delta, V\}, \tag{9}$$

with $\widetilde{J}_{\gamma,a}$ the phase-aligned response of Eq. (8).

3.2 Exact projective constraint map

The dynamics preserves the exchange-symmetric three-dimensional subspace. In a fixed symmetric basis we write the state as

$$\psi = (\psi_0, \psi_1, \psi_2)^T \in \mathbb{C}^3,$$

and eliminate the unobservable global phase by the projective coordinates

$$w_c = \frac{\psi_c}{\psi_0}, \qquad c = 1, 2, \tag{10}$$

valid on any region where the chart denominator ψ_0 excludes zero. For each error direction $a \in \{\Omega, \Delta, V\}$, the induced projective first response is

$$\dot{w}_{a,c} = \frac{\dot{\psi}_{a,c}\, \psi_0 - \psi_c\, \dot{\psi}_{a,0}}{\psi_0^2}, \qquad c = 1, 2, \tag{11}$$

where $\dot{\psi}_a = \partial_{\epsilon_a}\psi\big|_{\epsilon=0}$. The exact constraint map is then

$$F(\phi) = \begin{pmatrix} \Re(w - w_{\mathrm{ref}}) \\ \Im(w - w_{\mathrm{ref}}) \\ \Re(\dot{w}_\Omega - \dot{w}_{\Omega,\mathrm{ref}}) \\ \Im(\dot{w}_\Omega - \dot{w}_{\Omega,\mathrm{ref}}) \\ \Re(\dot{w}_\Delta - \dot{w}_{\Delta,\mathrm{ref}}) \\ \Im(\dot{w}_\Delta - \dot{w}_{\Delta,\mathrm{ref}}) \\ \Re(\dot{w}_V - \dot{w}_{V,\mathrm{ref}}) \\ \Im(\dot{w}_V - \dot{w}_{V,\mathrm{ref}}) \end{pmatrix} \in \mathbb{R}^{16}. \tag{12}$$

The exact matching used below is therefore exact projective output-state and first-projective-response matching: it is exact after elimination of the global phase, rather than a componentwise equality of raw state vectors. The numerical optimizer retains the full phase-aligned complex output state and the three full phase-aligned complex state tangents; redundant real components are retained in the residual vector, while singular-value decomposition determines the independent rank.

3.3 Quadratic degeneracy under first-response matching

The next lemma is model-independent. It upgrades the empirical agreement of quadratic loss coefficients to an exact statement: projective first-response matching forces two families to share the same quadratic loss form, so their symmetrized losses can first differ at fourth order.

Lemma 1 (Quadratic degeneracy under projective first-response matching)**.** *Let $\psi_\alpha(\epsilon)$ and $\psi_\beta(\epsilon)$ be normalized analytic state families in a finite-dimensional Hilbert space such that*

$$[\psi_\alpha(0)] = [\psi_\beta(0)] = [\psi_0]$$

projectively, with identical horizontal first derivatives at the origin, $h_{\alpha,a} = h_{\beta,a}$, where

$$h_{\gamma,a} = P_\perp \frac{\partial}{\partial \epsilon_a} |\psi_\gamma(\epsilon)\rangle\bigg|_{\epsilon=0}, \qquad P_\perp = I - |\psi_0\rangle\langle\psi_0|.$$

Then

$$\big(1-|\langle\psi_0|\psi_\alpha(\epsilon)\rangle|^2\big)-\big(1-|\langle\psi_0|\psi_\beta(\epsilon)\rangle|^2\big)=\mathcal{O}(\|\epsilon\|^3),$$

and for the symmetrized losses of Eq. (13)*,*

$$S_\alpha(t,v)-S_\beta(t,v)=\mathcal{O}(t^4).$$

Proof. The infidelity is invariant under a global phase, so after phase alignment we may take $\psi_\alpha(0)=\psi_\beta(0)=\psi_0$ and use the common target ψ_0. Write $f_\gamma(\epsilon)=1-|\langle\psi_0|\psi_\gamma(\epsilon)\rangle|^2$ and expand

$$\langle\psi_0|\psi_\gamma(\epsilon)\rangle=1+\epsilon_a A_a+\tfrac{1}{2}\epsilon_a\epsilon_b A_{ab}+\mathcal{O}(\|\epsilon\|^3),$$

with $A_a=\langle\psi_0|\partial_a\psi_\gamma\rangle$ and $A_{ab}=\langle\psi_0|\partial_a\partial_b\psi_\gamma\rangle$ evaluated at $\epsilon=0$; repeated indices are summed. Differentiating the normalization $\langle\psi_\gamma|\psi_\gamma\rangle=1$ gives $\Re A_a=0$ and $\Re A_{ab}=-\Re\langle\partial_a\psi_\gamma|\partial_b\psi_\gamma\rangle$. The linear term of f_γ therefore vanishes, and its quadratic term is the Fubini–Study form of the horizontal derivatives,

$$f_\gamma(\epsilon)=g_{ab}\,\epsilon_a\epsilon_b+\mathcal{O}(\|\epsilon\|^3),\qquad g_{ab}=\Re\langle h_{\gamma,a}|h_{\gamma,b}\rangle,$$

since $h_{\gamma,a}=\partial_a\psi_\gamma-A_a\psi_0$. Identical horizontal first derivatives give $g^\alpha_{ab}=g^\beta_{ab}$, hence $f_\alpha-f_\beta=\mathcal{O}(\|\epsilon\|^3)$. Symmetrization in t cancels the odd-order terms, so the difference of the symmetric responses starts at order t^4. □

3.4 Dimension count and generation

The control has 24 phase variables. At the reference solution, the numerical constraint Jacobian has rank

$$\operatorname{rank} DF(\gamma_{\text{ref}})=16,$$

leaving an eight-dimensional numerical null space that organizes the candidate implementation fibre. Candidate paths are generated by:

1. drawing a seeded direction in the numerical null space;
2. applying a finite displacement;
3. projecting back to the constraint set by nonlinear least squares;
4. rejecting candidates whose constraint residual or pairwise phase distance fails a predeclared gate.

No held-out finite-error outcome is evaluated during candidate generation. The maximum constraint residual in the reported runs is of order 10^{-11}.

Remark 1 (Global fibre versus local certified roots)*.* The numerical rank and residual establish a highly accurate computational fibre, suggesting a locally eight-dimensional numerical implementation fibre, not the exact existence of a global eight-dimensional solution manifold. A complete global fibre theorem would require a square transverse formulation and a global interval-Newton or Krawczyk proof. The present work instead establishes 12 locally unique roots exactly matched in projective output state and first projective response via strict Krawczyk inclusions in transverse boxes. The global fibre structure remains open.

Figure 1 summarizes the local mechanism. The drawing is schematic: it represents one local chart and does not assert that the twelve certified roots belong to one globally connected exact fibre. The geometric object used below is the quartic response tensor of the loss, not a curvature tensor of a chosen connection or metric.

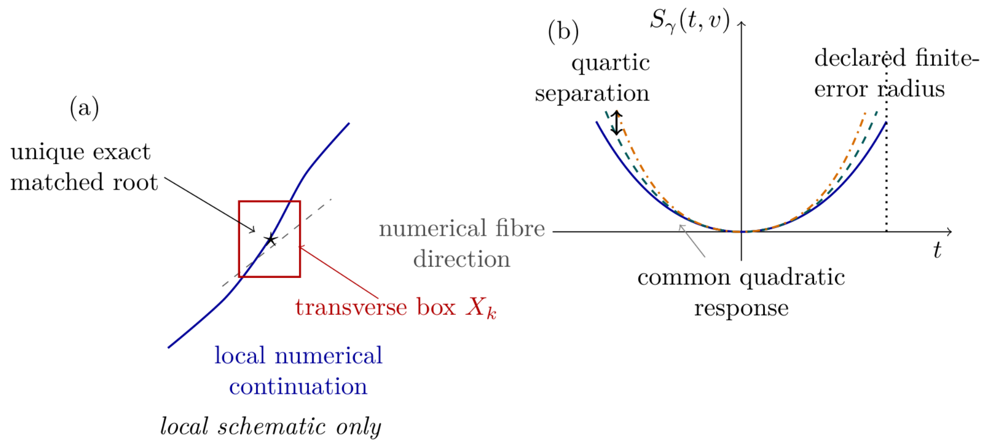

Figure 1: Schematic of local response matching and quartic separation. (a) In a local control-space chart, the projective output-state and first-response constraints define numerical tangent directions; the formal certificate establishes a unique exact matched root only inside each declared transverse box and does not assert a global connected fibre. (b) Exact projective first-response matching makes the quadratic loss form common. Symmetrized finite-error responses can therefore first separate through their quartic coefficients. The tensor $\mathcal{A}^{(4)}$ is a fourth-order response susceptibility, not a Riemannian curvature tensor.

4 Prospective fourth-order response prediction

4.1 Symmetric local response

Introduce normalized error coordinates $z \in \mathbb{R}^3$ through

$$\epsilon = Dz, \qquad D = \mathrm{diag}(0.06, 0.04, 0.05).$$

For a direction $v \in \mathbb{R}^3$, define the symmetric response

$$S_\gamma(t,v) = \frac{\mathcal{I}_\gamma^{\mathrm{loc}}(tDv) + \mathcal{I}_\gamma^{\mathrm{loc}}(-tDv)}{2}. \tag{13}$$

Analyticity of the finite matrix exponential gives

$$S_\gamma(t,v) = q_{2,\gamma}(v)t^2 + q_{4,\gamma}(v)t^4 + q_{6,\gamma}(v)t^6 + \cdots . \tag{14}$$

Odd powers cancel by construction.

By Lemma 1, exact projective first-response matching forces the quadratic loss forms of matched controls to coincide exactly, so the quartic term is the first possible symmetric discriminator. The numerical cohorts here are matched only within tolerance; consistently, the relative spread of the averaged quadratic coefficient across the validation cohorts is approximately 10^{-7}. The first useful discriminator is therefore the quartic coefficient.

4.2 Prospective quartic ranking

The quartic rule developed here is a *geometric predictor*, not a universal theorem: it is gap-dependent and certifies only a subset of pairwise comparisons, a boundary made precise in Sections 5–6.

4.2.1 Discovery and validation separation

The initial four named controls were used only for discovery and orientation. A twelve-path cohort was then used to compare candidate formulas. The smallest sufficient rule on that training cohort was

$$\text{smaller } \mathcal{G}_4 \implies \text{smaller mean finite-error infidelity.}$$

The addition of the sixth-order coefficient did not improve the training rank correlation, so the quartic rule was frozen.

For the prospective audit, the coefficients q_2, q_4, q_6 were estimated by a frozen least-squares fit of

$$S_\gamma(t, e_a) = q_{2,a}t^2 + q_{4,a}t^4 + q_{6,a}t^6$$

at

$$t \in \left\{\frac{1}{32}, \frac{1}{24}, \frac{1}{20}, \frac{1}{16}, \frac{1}{14}, \frac{1}{12}, \frac{1}{10}, \frac{1}{8}\right\}.$$

The tensor audit below is computationally distinct: its directional derivatives are evaluated directly at $t = 0$ by block-matrix exponentials, rather than by fitting nonzero-radius samples.

A new twenty-path cohort was generated with a different seed. For each candidate, a ranking certificate containing phases, local coefficients, and predicted order was written and SHA-256 hashed before the held-out outcome evaluator was enabled. The predeclared primary gate was

$$\rho_{\text{Spearman}} \geq 0.95,$$

together with correct top-path identification.

4.2.2 Quartic validation result

The twenty-path Colab run gave

$$\rho_{\text{Spearman}}\big(-\mathcal{G}_4, -\bar{\mathcal{I}}\big) = 0.996992. \tag{15}$$

This prospective rank correlation is empirical evidence on a separately generated held-out cohort. It is not used in the proof of the exact-root ordering theorem and does not establish a universal fourth-order ranking law. Because the quartic rule, ranking direction, acceptance threshold, and ranking certificate were frozen before held-out finite-error evaluation, the reported correlation is not an in-sample fit; nevertheless, it remains cohort-specific prospective evidence rather than a theorem. The predicted and observed rankings differed by two adjacent transpositions, corresponding to $\sum_i d_i^2 = 4$; this is a descriptive rank diagnostic for the frozen seeded cohort, and no statistical-population inference is claimed. The predicted and observed top path agreed. Two local coefficient-extraction windows had rank correlation one, and the maximum local fit residual was 1.48×10^{-10}.

Subsequent cohorts show that $\mathcal{G}_4$ is not universally sufficient: depending on the generated matched paths, the prospective quartic correlation varies appreciably across seeded cohorts. This variation motivated the bounded higher-order analysis rather than post hoc replacement of the quartic formula.

5 Covariant quartic response geometry

5.1 Quartic tensor

The object recovered in this section is a quartic susceptibility of the endpoint loss—a component of the fourth jet of the endpoint map composed with infidelity—not a Riemannian curvature or an instance of the quantum geometric tensor [19]. There exists a symmetric fourth-order tensor $\mathcal{A}^{(4)}_\gamma$ such that

$$q_{4,\gamma}(v) = \mathcal{A}^{(4)}_{\gamma,ijkl}\, v_i v_j v_k v_l. \tag{16}$$

In three dimensions a symmetric fourth-order tensor has $\binom{3+4-1}{4} = 15$ independent components. The tensor $\mathcal{A}^{(4)}_\gamma$ is reconstructed from the fourth jet of the path-local loss $\mathcal{I}^{\mathrm{loc}}_\gamma$: we recover its components from 36 seeded directions, whose directional Taylor coefficients are computed directly at $t = 0$ through order six by block-matrix exponentials, rather than by fitting nonzero-radius samples.

Let

$$z \sim \mathrm{Unif}\{\pm e_\Omega, \pm e_\Delta, \pm e_V\}, \qquad \epsilon = Dz,$$

so that the physical error scales $0.06^4, 0.04^4, 0.05^4$ are carried by D and absorbed into the tensor components. For the six-axis law in Eq. (5), define the fourth moment

$$M^{(4)}_{ijkl} = \mathbb{E}[z_i z_j z_k z_l]\,.$$

The scalar quartic predictor is

$$\mathcal{G}_4(\gamma; M^{(4)}) = \mathcal{A}^{(4)}_{\gamma,ijkl} M^{(4)}_{ijkl}. \tag{17}$$

For the equally weighted signed axes,

$$\mathcal{G}_4 = \frac{1}{3}\Big(\mathcal{A}^{(4)}_{\Omega\Omega\Omega\Omega} + \mathcal{A}^{(4)}_{\Delta\Delta\Delta\Delta} + \mathcal{A}^{(4)}_{VVVV}\Big).$$

5.2 Coordinate covariance

Proposition 1 (Covariance and invariant contraction)**.** *Let $z = Ry$ for an invertible real matrix R. Then*

$$\mathcal{A}^{(4)}_{y,abcd} = \mathcal{A}^{(4)}_{z,ijkl}\, R_{ia} R_{jb} R_{kc} R_{ld}, \tag{18}$$

$$M^{(4)}_{y,abcd} = (R^{-1})_{ai}(R^{-1})_{bj}(R^{-1})_{ck}(R^{-1})_{dl}\, M^{(4)}_{z,ijkl}, \tag{19}$$

and

$$\mathcal{A}^{(4)}_y : M^{(4)}_y = \mathcal{A}^{(4)}_z : M^{(4)}_z. \tag{20}$$

Proof. Equation (18) follows by substituting $z = Ry$ into the quartic form. Equation (19) follows from $y = R^{-1}z$. Contracting all four indices cancels each R with the corresponding R^{-1}, proving Eq. (20). $\square$

This proposition is algebraic. The numerical audit additionally recomputes the tensor directly in the transformed coordinates. For a nonorthogonal map of condition number approximately 1.82, the direct-refit relative error is 1.08×10^{-14} and the maximum invariant-contraction relative error is 2.36×10^{-15}.

The covariant object is not the scalar $\mathcal{G}_4$ in isolation. It is the fourth-order response tensor $\mathcal{A}^{(4)}$ together with the physical fourth noise moment $M^{(4)}$. Their contraction is independent of a linear reparameterization of error coordinates. This avoids interpreting a coordinate-specific axis average as an intrinsic curvature.

Establishing a relation of this susceptibility to a canonical connection or curvature on a control quotient would require additional structure.

For a noise moment M, the low-cost screening problem is

$$\gamma^\star \in \arg\min_{\gamma \in \mathcal{F}_\eta}\ \mathcal{A}^{(4)}_\gamma : M^{(4)}.$$

Pairs that fail the quartic gap test can be escalated adaptively to sixth, eighth, or higher jet order until their intervals separate. This suggests a certified branch-and-bound control selector: compute low-order local tensors for all candidates, refine only ambiguous pairs, and reserve direct finite-error simulation or hardware evaluation for model validation rather than exhaustive ranking.

The gap-dependent certification criterion used by the implementation retains the residual constant and quadratic terms explicitly. The certified quantity is the six-axis averaged reference loss at radius t,

$$\mathcal{I}_k^{\mathrm{ref}}(t) = \frac{1}{6} \sum_{a\in\{\Omega,\Delta,V\}} \sum_{\sigma=\pm1} \mathcal{I}_k^{\mathrm{ref}}(\sigma t D e_a). \tag{21}$$

For each path k write the truncated local model and its enclosure as

$$P_k(t) = C_{0,k} + C_{2,k}t^2 + \mathcal{G}_{4,k}t^4, \qquad \mathcal{I}_k^{\mathrm{ref}}(t) \in P_k(t) + [-B_k(t),\, B_k(t)]. \tag{22}$$

If

$$|P_a(t) - P_b(t)| > B_a(t) + B_b(t), \tag{23}$$

then the sign of $\mathcal{I}_a^{\mathrm{ref}}(t) - \mathcal{I}_b^{\mathrm{ref}}(t)$ is strictly certified. Only when $C_{0,a} = C_{0,b}$ and $C_{2,a} = C_{2,b}$ does this reduce to the simplified quartic-gap form

$$|\mathcal{G}_{4,a} - \mathcal{G}_{4,b}|\, |t|^4 > B_a(t) + B_b(t).$$

The implementation folds the residual C_0 and C_2 differences into the correction intervals, so they are not omitted from the certificate.

6 Computer-assisted certification

6.1 Krawczyk root theorem

The numerical paths satisfy the endpoint and first-order matching constraints to order 10^{-11}. To move from numerical candidates to exact mathematical roots, we use the projective constraint map $F : \mathbb{R}^{24} \to \mathbb{R}^{16}$ of Eq. (12) and decompose the phase space into 16 transverse variables x and 8 fibre variables y. For each declared numerical path γ_k, we fix its fibre coordinates y_k and consider the reduced square map

$$F_k(x) = F(x, y_k).$$

For a box $X \subset \mathbb{R}^{16}$ with centre x_0, the Krawczyk operator is

$$K(X) = x_0 - CF(x_0) + \big(I - C[DF(X)]\big)(X - x_0), \tag{24}$$

where C is a frozen nonsingular numerical preconditioner (an approximate inverse of the midpoint Jacobian) and $[DF(X)]$ is an interval enclosure of the Jacobian over X. In centered coordinates $x_0 = 0$, as used by the implementation, this reduces to

$$K(X) = -CF(0) + \big(I - C[DF(X)]\big)X.$$

The strict inclusion $K(X) \subset \mathrm{int}(X)$ then implies that F_k has a unique zero inside X [20, 9]. The projective chart denominator ψ_0 of Eq. (10) is formally verified to exclude zero on every accepted box, ensuring the coordinate chart is valid throughout the Krawczyk computation.

Theorem 1 (Krawczyk inclusions for 12 declared controls)**.** *For each of the twelve declared controls, there exists a transverse phase box*

$$X_k \subset \mathbb{R}^{16}$$

such that the Krawczyk operator satisfies

$$K_k(X_k) \subset \mathrm{int}(X_k).$$

Consequently, each box X_k contains a locally unique exact root x_k^* of $F_k(x) = 0$, i.e., a control $\gamma_k^* = (x_k^*, y_k)$ that is exactly matched in projective output state and first projective response to the reference.

Computer-assisted proof. A numerical approximate inverse of the midpoint Jacobian is used as the point preconditioner. The Jacobian over each full box is then enclosed with 192-bit Arb arithmetic, and the resulting Krawczyk image is verified to lie strictly inside the box. The projective chart denominator is formally verified to exclude zero on every accepted box. For all twelve boxes, the inclusion $K(X) \subset \mathrm{int}(X)$ is verified. The SHA-256 protocol and certificate identifiers of this run are recorded in Appendix B. By the Krawczyk theorem [20, 9, 10], each box contains a locally unique root. □

Remark 2 (Scope of the Krawczyk inclusion theorem)*.* The theorem establishes local uniqueness inside each declared transverse box. It does not certify global uniqueness on the entire fibre, nor does it prove that every point on the numerical fibre has an exact representative. The global fibre structure remains open.

6.2 Direct finite-error ordering theorem

The following theorem is the main result of this work. Its proof uses only the certified Krawczyk boxes and direct outward-rounded propagation of the finite-error functional. No Taylor truncation, Cauchy extraction, alias enclosure, or local-jet reconstruction is used.

Theorem 2 (Exact-root finite-error ordering)**.** *For each of the twelve declared controls, let X_k be the transverse phase box satisfying a strict Krawczyk inclusion from Theorem 1. Direct outward-rounded Arb propagation of X_k through the declared six-point finite-error functional produces twelve performance intervals that are pairwise disjoint and totally ordered according to the pre-outcome ordering frozen for the independent twelve-path formal cohort. Therefore the locally unique roots exactly matched in projective output state and first projective response inside the boxes obey the complete frozen finite-error ordering.*

Computer-assisted proof. By Theorem 1, the locally unique projectively matched control γ_k^* belongs to the certified phase box X_k. For every box X_k, the six declared finite-error Hamiltonians are evaluated directly using 192-bit outward-rounded Arb/ACB arithmetic. Every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. Denote the resulting real performance interval by B_k. Therefore

$$\bar{\mathcal{I}}(\gamma_k^*) \in B_k.$$

For every frozen ordered pair $a \prec b$, direct interval comparison verifies

$$\sup B_a < \inf B_b.$$

All $\binom{12}{2} = 66$ comparisons pass and have the orientation specified by the frozen ordering certificate. Hence

$$\bar{\mathcal{I}}(\gamma_a^*) < \bar{\mathcal{I}}(\gamma_b^*)$$

for every frozen ordered pair. No Taylor truncation or local-jet reconstruction is used in this direct certification. □

The predicted ordering was fixed by the predecessor zero-point ordering certificate before the exact-root direct finite-error propagation was evaluated. The final exact-root certificate, containing the formal direct intervals and all pairwise comparisons, was written and hashed after the propagation. The SHA-256 protocol and certificate identifiers of this run are recorded in Appendix B.

Remark 3 (Scope of Theorem 2). The theorem is conditional on the serialized-decimal finite Hamiltonian model, the declared transverse boxes, and the six-axis mean error functional. It does not certify model discrepancy, global fibre topology, many-body scaling, or hardware. It also does not prove a global eight-dimensional exact fibre theorem; it proves finite-error ordering for the locally unique exact roots inside the declared boxes.

Corollary 1 (Ordering stability under bounded loss discrepancy)**.** *Let B_k denote the certified direct performance interval for the kth exact root, and define the minimum certified ordering margin*

$$\Delta_{\min} = \min_{a \prec b}\bigl(\inf B_b - \sup B_a\bigr) > 0,$$

where $a \prec b$ ranges over the frozen ordered path pairs. Suppose an additional physical effect changes the evaluated loss of every fixed certified control by at most η:

$$\bigl|\tilde{\bar{\mathcal{I}}}(\gamma_k^*) - \bar{\mathcal{I}}(\gamma_k^*)\bigr| \le \eta \quad \textit{for every } k.$$

If

$$2\eta < \Delta_{\min},$$

then the complete frozen ordering of the twelve fixed certified controls is preserved under the perturbed loss.

Proof. For every frozen ordered pair $a \prec b$,

$$\tilde{\bar{\mathcal{I}}}(\gamma_a^*) \le \sup B_a + \eta < \inf B_b - \eta \le \tilde{\bar{\mathcal{I}}}(\gamma_b^*),$$

where the strict middle inequality follows from $2\eta < \Delta_{\min}$. □

For the frozen twelve-root certificate, the minimum formally certified pairwise margin, computed from the outward-rounded endpoints of the frozen direct-certificate intervals, is

$$\Delta_{\min} = 2.50 \times 10^{-5},$$

attained by the pair pv08 and pv11.

Remark 4. Corollary 1 is a conditional stability statement for the losses evaluated at the same certified controls. It does not prove that an added Lindblad term, waveform filter, calibration drift, or other model perturbation satisfies the required bound. It also does not certify that the controls remain exact response-matched roots of the perturbed model. Proving root continuation under model perturbation would require a parameterized Krawczyk or validated implicit-function argument.

Remark 5 (Integrity check). In all reported runs, every ordinary held-out floating-point outcome is verified to lie inside its corresponding formal ball. This serves as an integrity diagnostic and is not part of the theorem proof.

6.3 Order-thirty mechanism certificate

The following proposition provides a separate, computationally distinct mechanism certificate using the zero-point jet. Unlike the main theorem, it uses Taylor coefficients, Cauchy extraction, alias enclosures, and Dyson tail bounds. It is not part of the main theorem proof.

The floating-point precursor obtains the order-thirty jet by block-matrix exponentials. The formal phase-box certificate used in Proposition 2 instead extracts the coefficients with a 64-point outward-rounded Cauchy transform.

Proposition 2 (Order-thirty mechanism certificate)**.** *Propagation of the zero-point jet through order thirty over the twelve Krawczyk phase boxes, including the formal Cauchy-alias and analytic order-32 tail enclosures, certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified.*

Computer-assisted proof. For path k and physical error axis a, define the analytic transition amplitude

$$g_{k,a}(\lambda) = \langle \psi_{\mathrm{ref}} | \psi_{\gamma_k}(\lambda D e_a) \rangle = \sum_{n \geq 0} g_{k,a,n}\, \lambda^n.$$

Its coefficients are extracted by a 64-point outward-rounded Cauchy discrete Fourier transform on the complex unit circle:

$$\widehat{g}_n = \frac{1}{64} \sum_{k=0}^{63} g(\zeta_k)\, \zeta_k^{-n}, \qquad \zeta_k = e^{2\pi i k/64}. \tag{25}$$

The discrete transform aliases coefficients $g_{n+64\ell}$. A path-uniform Dyson bound [8] encloses this alias. The implemented enclosure is 10^{-40}, and the formal gate verifies that

$$\frac{e^{K_a} K_a^{64}}{64!} < 10^{-40}$$

for every physical error axis, where the dependence is parameterized as

$$H_a(s; \lambda_a) = H_0(s) + \lambda_a H_{1,a}(s),$$

with s the physical time and λ_a the dimensionless error parameter along axis a, and

$$K_a = \int_0^T \|H_{1,a}(s)\|_2\, ds = \int_0^T \left\| \frac{\partial H_a(s; \lambda_a)}{\partial \lambda_a} \bigg|_{\lambda_a = 0} \right\|_2 ds$$

is the integrated spectral norm of the Hamiltonian perturbation along axis a. Fidelity coefficients are formed by ball convolution,

$$F_n = \sum_{r=0}^{n} g_r\, \bar{g}_{n-r},$$

and the infidelity coefficients follow from $1 - F$. The analytic even Dyson tail after order thirty is enclosed by 2×10^{-11}.

Propagating the resulting order-thirty jet over the twelve Krawczyk phase boxes yields performance intervals. Pairwise comparison of the resulting order-thirty intervals certifies 52/66 frozen path pairs. Every certified orientation agrees with the frozen order, and no reversed pair is certified. The unresolved pairs arise from dependency and wrapping in the phase-box propagation of the jet enclosure. They are not reversed physical comparisons and do not indicate an omitted-tail failure. □

Table 1: Representative prospective and formal audits. "Rank diagnostic" denotes the Spearman correlation of the corresponding ranking stage; covariance itself is verified by the 10^{-14}–10^{-15} relative residuals collected in Appendix B. "Certified result" states the certified object explicitly, distinguishing certified roots from certified unordered pairs. Candidate sets differ between stages and between numerical environments; every stage freezes its protocol and ranking before held-out evaluation. The direct exact-root ordering is the main theorem.

Audit	Paths	Rank diagnostic	Certified result	Logical role
Prospective $\mathcal{G}_4$ ranking	20	$\rho = 0.996992$	–	prospective evidence
Covariant $\mathcal{G}_4$ contraction ranking	12	$\rho = 0.965035$	–	covariance/ranking audit
Formal quartic-only bound	12	–	34/66 pairs	low-order boundary
Frozen numerical-point order-30	12	$\rho = 1$	66/66 pairs	diagnostic
Exact-root Krawczyk inclusion	12	–	12/12 roots	existence/uniqueness
Exact-root order-30 jet	12	–	52/66 pairs	mechanism certificate
Exact-root direct Arb propagation	12	$\rho = 1$	66/66 pairs	main theorem

7 Results

Table 1 collects the audits in the fixed hierarchy of the paper: the 66/66 direct exact-root ordering theorem (main result), supported by the 12/12 Krawczyk local uniqueness certificates; the 52/66 order-thirty mechanism certificate; the 34/66 quartic-only boundary; and the prospective $\rho = 0.996992$ evidence on the separate cohort. A frozen-point diagnostic sharpens the picture: for the twelve serialized numerical phase points, before propagation over the Krawczyk root boxes, the order-thirty jet and its analytic tail enclosure certify all 66/66 pairwise comparisons. This pointwise diagnostic is witnessed by the proof objects in `results/14_order30/`; it is not an exact-response-root certificate. Propagation over the certified root boxes subsequently resolves 52/66 pairs with zero reversals (Proposition 2). Numerical integrity diagnostics from representative runs are collected in Appendix B.

Figure 2 displays the two data-level statements behind the title: the prospective ranking on the held-out cohort and the certified exact-root intervals of the main theorem.

8 Discussion

8.1 Why the quartic term is useful but incomplete

First-order matching makes the quadratic loss nearly common—exactly common at exact matched roots by Lemma 1. This promotes the quartic term to the first visible discriminator. It explains why $\mathcal{G}_4$ often gives almost perfect rankings. However, if two quartic coefficients are close, the sixth and higher terms can reverse their order. The formal intervals expose this distinction directly: 34 pairs are quartic-certified, while the remaining pairs require higher jet data or direct propagation.

This is preferable to declaring the quartic rule either universally true or false. The correct statement is gap-dependent as in Eq. (23).

8.2 Minimal model versus scalable certification

The role of the two-atom model is to isolate the response-matching mechanism in the smallest interacting Rydberg system, not to approximate a many-body processor. The present certificate shows that residual finite-error ordering survives exact projective state and complete first-response matching without being attributable to optimizer residuals. Turning this result into a

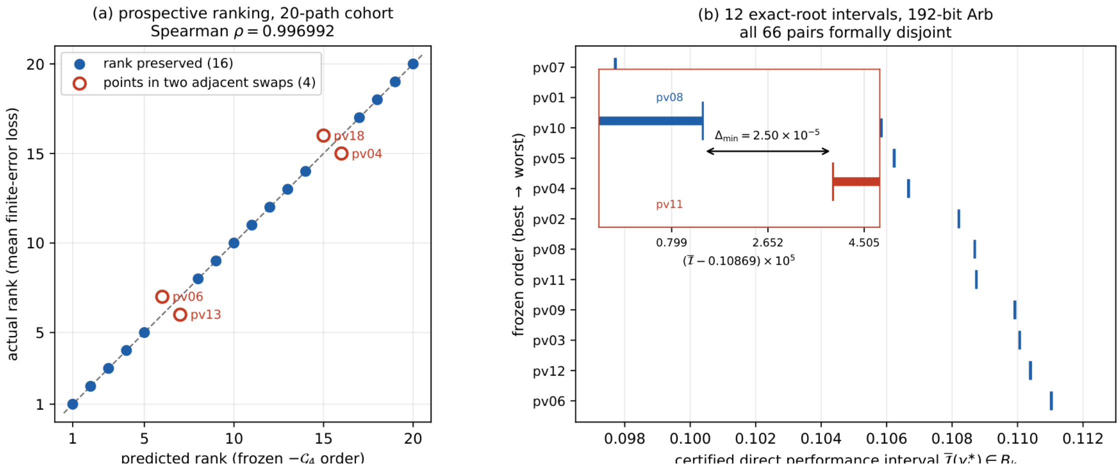

Figure 2: Direct visual evidence for the two claims in the title. (a) Prospective prediction: $-\mathcal{G}_4$ predicted versus actual ranks on the held-out twenty-path cohort. The frozen order reproduces the actual mean-loss ranking up to two adjacent transpositions (marked), giving Spearman $\rho = 0.996992$. This panel reports prospective empirical evidence; it is not a certificate. (b) Exact-root certification: the twelve certified direct performance intervals B_k from 192-bit outward-rounded Arb propagation, drawn to scale in the frozen order. All 66 unordered pairs are formally disjoint; the tightest pair, pv08–pv11, has the certified margin $\Delta_{\min} = 2.50 \times 10^{-5}$ of Corollary 1 (inset: magnification of that pair).

many-body compilation method requires a separate scalability programme, including structured state representations and dependency-preserving validated propagation. The current theorem should therefore be read as a minimal rigorous existence-and-ordering result rather than as a scalable hardware-design algorithm.

8.3 Order-30 mechanism versus direct propagation

The order-30 jet propagated over the Krawczyk boxes certifies 52/66 pairs with zero reversals (Proposition 2). The remaining 14 pairs are not physically reversed—they are unresolved due to dependency and wrapping in the phase-box propagation of the jet enclosure. Direct Arb propagation of the same boxes certifies all 66 pairs (Theorem 2), demonstrating that the unresolved cases are an artefact of the jet propagation and interval wrapping, not a physical failure of the frozen ordering. The 52/66 mechanism certificate and the 66/66 direct certificate arise from different propagation levels and should not be conflated: the latter is the main theorem.

The content is quantitative rather than existential: an analytic finite-dimensional propagator is determined by its local germ, but here a fixed finite order, 30, resolves 52/66 comparisons with an explicitly bounded tail, and the unresolved cases arise from interval wrapping rather than truncation.

The particular ordering of the twelve serialized controls is not presented as a universal physical law. The reusable contribution is the certificate architecture: transverse exact-root isolation, formal propagation of the resulting root boxes, and explicit separation between low-order mechanism certificates and direct finite-error theorems.

9 Limitations and next theorem

The present work has five explicit limitations.

1. **Global fibre structure.** Twelve locally unique roots exactly matched in projective output state and first projective response are certified in transverse boxes. The global eight-dimensional solution manifold structure remains unproved.
2. **Finite-dimensional model.** Spontaneous emission, Doppler effects, laser phase noise, position fluctuations, leakage levels, and waveform filtering are absent. An open-system extension is not obtained by appending a dissipative curve to the present certificate. Exact matching of density matrices or quantum channels and their first responses defines a different constraint map, whose dimension generally scales with d^2 rather than d. It would require new matched controls, new root boxes, and validated Liouvillian propagation. Arbitrary dissipative perturbations need not preserve the present ordering, and no noise-model-independent ordering theorem is claimed here.
3. **Two atoms.** No many-body scaling theorem is claimed.
4. **Mean axial error law.** The primary result concerns the declared six-axis mean. Worst-case ranking is not supported by the present scalar.
5. **No QPU evidence.** The calculations are local and require no hardware access. They do not constitute physical-QPU evidence.

In short, the certificate is formal for the serialized finite-dimensional two-atom Hamiltonian, the declared transverse boxes, and the six-axis mean error functional; it does not certify model discrepancy, global fibre topology, many-body scaling, or hardware.

9.1 Constraint dimension and scalability

The present dense interval implementation is not a scalable many-body compiler. Its dimensional growth can nevertheless be stated explicitly. Suppose the reachable complex state space has dimension d, and let m be the number of calibrated error coordinates. Matching one projective output state and its m complete projective first responses imposes

$$q = 2(m + 1)(d - 1)$$

real projective constraints. When the constraint Jacobian has rank q, a square Krawczyk chart uses q transverse control variables, while any remaining independent control variables parameterize numerical fibre directions.

Here $m = 3$ and the exchange-symmetric two-atom state space has $d = 3$, giving

$$q = 2(3 + 1)(3 - 1) = 16,$$

with $24 - 16 = 8$ numerical fibre directions. For a fully permutation-symmetric N-atom two-level model, $d = N + 1$ and the corresponding constraint count is $8N$. For a generic unsymmetrized model, $d = 2^N$ and the dense constraint count becomes $8(2^N - 1)$.

Thus exponential growth is not intrinsic to the Krawczyk operator alone; it arises through the dimension of the reachable state representation and the cost of enclosing the associated propagators and Jacobians. In generic many-body settings, direct dense Arb/ACB propagation and naive interval Jacobians would also suffer rapidly increasing dependency and wrapping. Possible routes to larger systems include symmetry reduction, sparse or block-structured propagation, reduced-observable matching, tensor-network enclosures, Taylor or Chebyshev polynomial models with validated remainder bounds, interval subdivision, and parameter-dependent

Krawczyk charts. Taylor or Chebyshev polynomial models with validated remainder bounds may help reduce local propagation overestimation, but they do not by themselves remove the exponential growth of a generic many-body state representation. None of these extensions is implemented or certified in the present work.

The next mathematical step is a validated implicit-function theorem or a parameter-dependent Krawczyk atlas: treat the eight fibre coordinates as parameters and certify a unique transverse solution varying over a declared fibre neighbourhood. The target mathematical structure is

$$F(x, y) = 0, \qquad x \in \mathbb{R}^{16},\ y \in \mathbb{R}^{8},$$

and the goal is to prove that a unique transverse solution $x = x(y)$ exists and is regular over a declared y-region, not to prove a single global isolated root.

10 Reproducibility

The complete formal pipeline is frozen in a single standalone audit (v1.3). The twelve-path cohort, the transverse bases, and the point preconditioners are fixed; the common Krawczyk radius is 3×10^{-12}; and runtime metadata is excluded from every hashed proof object. Two complete formal executions produce byte-identical protocol, certificate, and report files, and the full audit was also rerun in a clean external Colab environment from the public v0.3.1 Git tag (commit `284974c9f6b952f4e114c8c5bdc9c2c299c4065c`) with all checks passing and no vendor account required. The SHA-256 certificate identifiers, numerical integrity diagnostics, script inventory, and clean-environment run records are collected in Appendices B and C.

11 Conclusion

The main result is a computer-assisted exact-root finite-error ordering theorem: on the frozen twelve-path cohort, strict Krawczyk inclusions certify twelve locally unique roots exactly matched in projective output state and first projective response, and direct 192-bit outward-rounded Arb propagation of the certified root boxes certifies the complete frozen ordering for all 66 unordered path pairs.

The mechanism behind the ordering is geometric. Exact projective first-response matching makes the quadratic loss form common, so the first possible discriminator is the quartic response tensor, whose covariant contraction with the noise moment organizes the comparison: it certifies 34/66 pairs alone, and a zero-point jet through order thirty certifies 52/66 with no reversals as a computationally independent mechanism certificate. On a separate twenty-path cohort, the frozen quartic contraction predicts the held-out ranking prospectively ($\rho = 0.996992$); this evidence is an application of the same structure, not a premise of the theorem.

The certificate is conditional on the declared serialized model and error functional. The next mathematical step is a parameterized Krawczyk or validated implicit-function continuation over the fibre coordinates; scalability, open-system, and hardware extensions each require new certificates, not extensions of the present one.

A Interval-arithmetic certificate construction

For completeness we collect the construction parameters of the formal certificates. All formal objects are computed in 192-bit outward-rounded Arb/ACB ball arithmetic [7]: every phase-dependent trigonometric factor, Hamiltonian matrix exponential, state propagation, overlap, infidelity, and six-point mean is enclosed. For each declared path, the Krawczyk computation uses a point preconditioner given by a numerical inverse of the midpoint Jacobian, a common transverse box radius of 3×10^{-12}, an interval enclosure of the Jacobian over the full box, and the strict inclusion test $K(X) \subset \mathrm{int}(X)$; the projective chart denominator ψ_0 is verified to exclude zero on every accepted box. The order-thirty mechanism certificate extracts Taylor coefficients by a 64-point outward-rounded Cauchy discrete Fourier transform; the discrete-transform alias is enclosed by the path-uniform Dyson bound $e^{K_a} K_a^{64}/64! < 10^{-40}$ on every physical error axis, and the analytic even Dyson tail after order thirty is enclosed by 2×10^{-11}. As an integrity diagnostic, every ordinary held-out floating-point outcome is verified to lie inside its corresponding formal ball; this diagnostic is not part of the theorem proofs.

B Certificate identifiers and integrity diagnostics

The frozen twelve-path cohort hash is

`9942af610c1e9499a10e989e4ae069e3`
`912d58eec4fd3ad1d3d39947c232c356`.

The deterministic protocol and certificate hashes are:

- Krawczyk protocol:
  `a19d01ec77a783be4e3858dd92ca080a`
  `a9fde2b09b77541b0e1ad7f392ff3124`
- Krawczyk certificate:
  `fc3ad3823597bd5281b7b9292f227f72`
  `3c2b3a9a8aa4e42c9f0353921b4d8fec`
- Exact-root ordering protocol:
  `f51ecfd2e4572f271e276cae18697005`
  `baa3f0ec3f3ab94f98f919cd50c3f844`
- Exact-root ordering certificate:
  `03c4db75cb80a149ce1a61e11c227039`
  `f9bd79c3f1304b3defa311a491ea2d1e`.

Table 2: Numerical integrity diagnostics from representative Colab runs.

Diagnostic	Value
Fourth-tensor independent components	15
Maximum tensor recovery relative residual	4.63×10^{-15}
Direct coordinate-refit relative error	1.08×10^{-14}
Invariant-contraction relative error	2.36×10^{-15}
Quadratic-coefficient relative spread	1.09×10^{-7}
Arb precision	192 bits
Cauchy points	64
Taylor order	30
Formal quartic pair coverage	$34/66 = 51.52\%$
Exact-root order-30 mechanism coverage	$52/66 = 78.79\%$
Exact-root direct Arb ordering coverage	$66/66 = 100\%$

C Scripts and clean-environment reproduction

The principal standalone scripts are:

```
pasqal_two_atom_G4_standalone_colab.py
pasqal_L3_L4_standalone_colab.py
pasqal_L4_order30_standalone_colab.py
pasqal_L4_formal_arb_standalone_colab.py
pasqal_L4_exact_fibre_krawczyk_standalone_colab_v1_2.py
pasqal_L4_exact_root_ordering_standalone_colab.py
pasqal_L4_reproducible_certificate_v1_3_colab.py
```

The last script produces all formal proof objects. The formal standalone script pins `python-flint==0.8.0`. NumPy and SciPy provide numerical diagnostics, while the v1.3 proof consumes the frozen decimal cohort and uses Arb/ACB for the formal interval certificate.

Two independent clean-environment Colab executions of the complete v1.3 audit, started from the public v0.3.1 tag, completed in 52.6 s and 107.9 s respectively; in both runs, dependency installation, repository SHA-256 verification, reference-artifact validation, regression tests, and the two-run byte-identical formal recomputation passed.

Code and Data Availability

All code, frozen inputs, formal certificates, and reproduction scripts are available at

https://github.com/papasop/quantum-control-geometry

The formal computational artifacts used in this work are fixed in release v0.3.1 at commit

`284974c9f6b952f4e114c8c5bdc9c2c299c4065c`.

The synchronized manuscript source is maintained on the repository's `main` branch. No PASQAL account or cloud credentials are required.

Funding

The author received no external funding for this work.

Competing Interests

The author declares no competing interests.

Author Contributions

Y.Y.N.Li conceived the study, developed the theoretical framework, implemented the numerical and formal audits, analyzed the results, and wrote the manuscript.

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